

13.3.2 I/O pin alternate function multiplexer and mapping

The device I/O pins are connected to on-board peripherals/modules through a multiplexer that allows only one peripheral alternate function (AF) connected to an I/O pin at a time. In this way, there is no conflict between peripherals available on the same I/O pin.

Each I/O pin has a multiplexer with up to 16 alternate function inputs (AF0 to AF15) that can be configured through the GPIOx_AFRL (for pin 0 to 7) and GPIOx_AFRH (for pin 8 to 15) registers:

- After reset, the multiplexer selection is alternate function 0 (AF0). The I/Os are configured in alternate function mode through GPIOx_MODER register.
- The specific alternate function assignments for each pin are detailed in the device datasheet.

In addition to this flexible I/O multiplexing architecture, each peripheral has alternate functions mapped onto different I/O pins to optimize the number of peripherals available in smaller packages.

To use an I/O in a given configuration, the user must proceed as follows:

- **Debug function:** after each device reset these pins are assigned as alternate function pins immediately usable by the debugger host.
- **GPIO:** configure the desired I/O as output, input, or analog in the GPIOx_MODER register.
- **Peripheral alternate function:**
 - Connect the I/O to the desired AFx in one of the GPIOx_AFRL or GPIOx_AFRH register.
 - Select the type, pull-up/pull-down, and output speed via the GPIOx_OTYPER, GPIOx_PUPDR and GPIOx_OSPEEDR registers respectively.
 - Configure the desired I/O as an alternate function in the GPIOx_MODER register.
- **Additional functions:**
 - For the ADC and DAC, configure the desired I/O in analog mode in the GPIOx_MODER register and configure the required function in the ADC and DAC registers.
 - For the additional functions like RTC, WKUPx, and oscillators, configure the required function in the related RTC, PWR, and RCC registers. These functions have priority over the configuration in the standard GPIO registers.

Refer to the “Alternate function mapping” table in the device datasheet for the detailed mapping of the alternate function I/O pins.

13.3.3 I/O port control registers

Each of the GPIO ports has four 32-bit memory-mapped control registers (GPIOx_MODER, GPIOx_OTYPER, GPIOx_OSPEEDR, GPIOx_PUPDR) to configure up to 16 I/Os. The GPIOx_MODER register is used to select the I/O mode (input, output, AF, analog). The GPIOx_OTYPER and GPIOx_OSPEEDR registers are used to select the output type (push-pull or open-drain) and speed. The GPIOx_PUPDR register is used to select the pull-up/pull-down whatever the I/O direction.

13.3.4 I/O port data registers

Each GPIO has two 16-bit memory-mapped data registers: input and output data registers (*GPIO port input data register (GPIOx_IDR) (x = A to I)* and *GPIO port output data register (GPIOx_ODR) (x = A to I)*).

GPIOx_ODR stores the data to be output, it is read/write accessible. The data entered through the I/Os are stored into the input data register (GPIOx_IDR), a read-only register.

13.3.5 I/O data bitwise handling

The bit set reset register (GPIOx_BSRR) is a 32-bit register that allows the application to set and reset each individual bit in the output data register (GPIOx_ODR). The bit set reset register has twice the size of GPIOx_ODR.

To each bit in GPIOx_ODR, correspond two control bits in GPIOx_BSRR: BS(i) and BR(i). When written to 1, BS(i) sets the corresponding ODR(i) bit. When written to 1, BR(i) resets the ODR(i) corresponding bit.

Writing any bit to 0 in GPIOx_BSRR does not have any effect on the corresponding bit in GPIOx_ODR. If there is an attempt to both set and reset a bit in GPIOx_BSRR, the set action takes priority.

Using the GPIOx_BSRR register to change the values of individual bits in GPIOx_ODR is a “one-shot” effect that does not lock the GPIOx_ODR bits. The GPIOx_ODR bits can always be accessed directly. The GPIOx_BSRR register provides a way of performing atomic bitwise handling.

There is no need for the software to disable interrupts when programming the GPIOx_ODR at bit level: one or more bits can be modified in a single atomic AHB write access.

13.3.6 GPIO locking mechanism

The GPIO control registers can be frozen by applying a specific write sequence to the GPIOx_LCKR register. The frozen registers are GPIOx_MODER, GPIOx_OTYPER, GPIOx_OSPEEDR, GPIOx_PUPDR, GPIOx_AFRL, GPIOx_AFRH and GPIOx_HSLVR.

To write the GPIOx_LCKR register, a specific write/read sequence must be applied. When the right LOCK sequence is applied to the bit 16 in this register, the value of LCKR[15:0] is used to lock the configuration of the I/Os (during the write sequence the LCKR[15:0] value must be the same). When the LOCK sequence is applied to a port bit, the value of the port bit can no longer be modified until the next MCU reset or peripheral reset. Each GPIOx_LCKR bit freezes the corresponding bit in the control registers (GPIOx_MODER, GPIOx_OTYPER, GPIOx_OSPEEDR, GPIOx_PUPDR, GPIOx_AFRL and GPIOx_AFRH).

The LOCK sequence can be performed only using a word (32-bit long) access to the GPIOx_LCKR register, because GPIOx_LCKR bit 16 must be set at the same time as the [15:0] bits.

13.3.7 I/O alternate function input/output

Two registers are provided to select one of the alternate function inputs/outputs available for each I/O. With these registers, the user can connect an alternate function to some other pin as required by the application.

This means that some peripheral functions are multiplexed on each GPIO using the GPIOx_AFRL and GPIOx_AFRH alternate function registers. The application can thus

select any one of the possible functions for each I/O. The AF selection signal being common to the alternate function input and alternate function output, a single channel is selected for the alternate function input/output of a given I/O.

To know which functions are multiplexed on each GPIO pin, refer to the device datasheet.

13.3.8 External interrupt/wake-up lines

All ports have external interrupt capability. To use external interrupt lines, the port can be configured in input, output, or alternate function mode (the port must not be configured in analog mode). Refer to [Section 18: Extended interrupts and event controller \(EXTI\)](#).

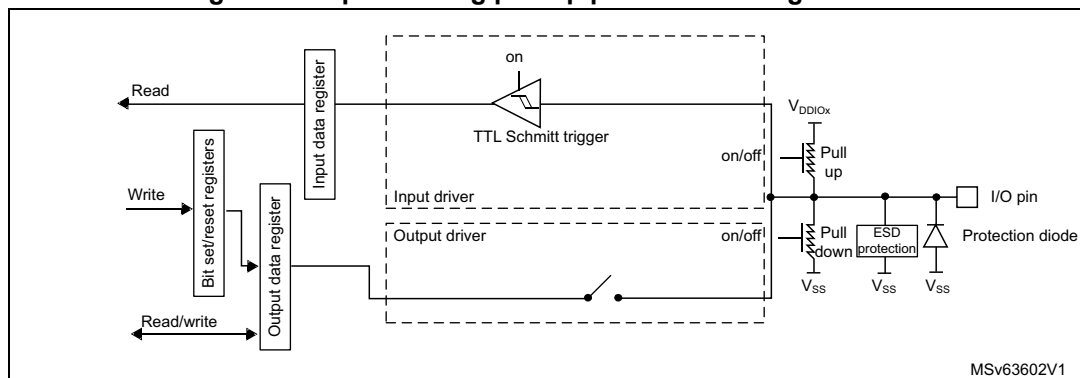
13.3.9 Input configuration

When the I/O port is programmed as input:

- The output buffer is disabled.
- The Schmitt trigger input is activated.
- The pull-up and pull-down resistors are activated depending on the value in the GPIOx_PUPDR register.
- The data present on the I/O pin are sampled into the input data register every AHB clock cycle.
- A read access to the input data register provides the I/O state.

[Figure 60](#) shows the input configuration of the I/O port bit.

Figure 60. Input floating/pull-up/pull-down configurations



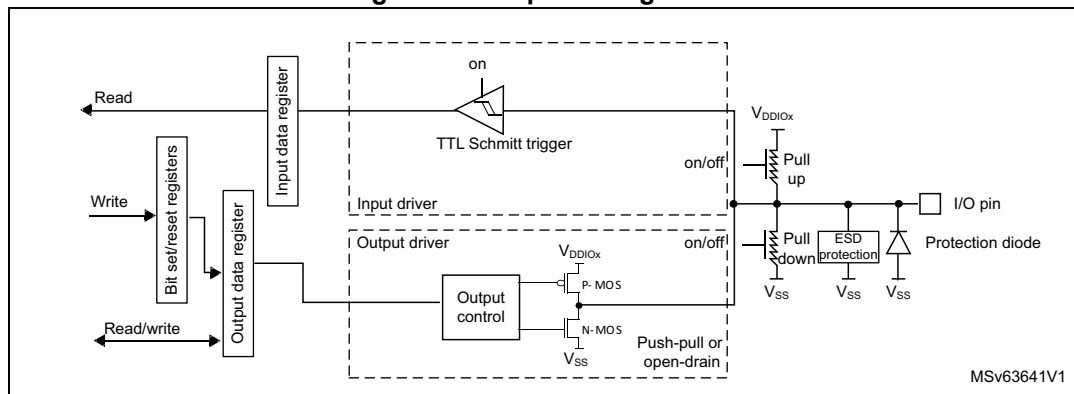
13.3.10 Output configuration

When the I/O port is programmed as output:

- The output buffer is enabled:
 - Open-drain mode: a 0 in the output register activates the N-MOS whereas a 1 in the output register leaves the port in high-Z (the P-MOS is never activated).
 - Push-pull mode: a 0 in the output register activates the N-MOS whereas a 1 in the output register activates the P-MOS.
- The Schmitt trigger input is activated.
- The pull-up and pull-down resistors are activated depending on the value in the GPIOx_PUPDR register.
- The data present on the I/O pin are sampled into the input data register every AHB clock cycle.
- A read access to the input data register gets the I/O state.
- A read access to the output data register gets the last written value.

Figure 61 shows the output configuration of the I/O port bit.

Figure 61. Output configuration



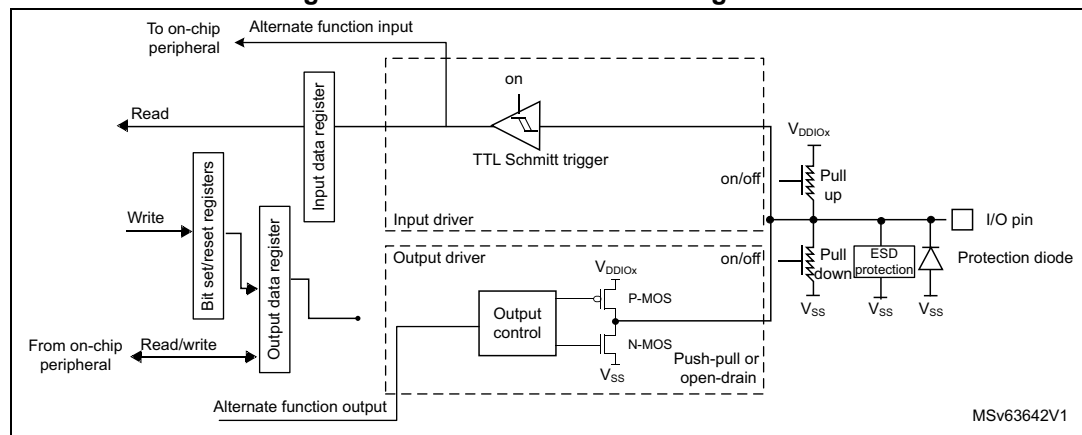
13.3.11 Alternate function configuration

When the I/O port is programmed as alternate function:

- The output buffer can be configured in open-drain or push-pull mode.
- The output buffer is driven by the signals coming from the peripheral (transmitter enable and data).
- The Schmitt trigger input is activated.
- The weak pull-up and pull-down resistors are activated or not depending on the value in the GPIOx_PUPDR register.
- The data present on the I/O pin are sampled into the input data register every AHB clock cycle.
- A read access to the input data register gets the I/O state.

Figure 62 shows the alternate function configuration of the I/O port bit.

Figure 62. Alternate function configuration



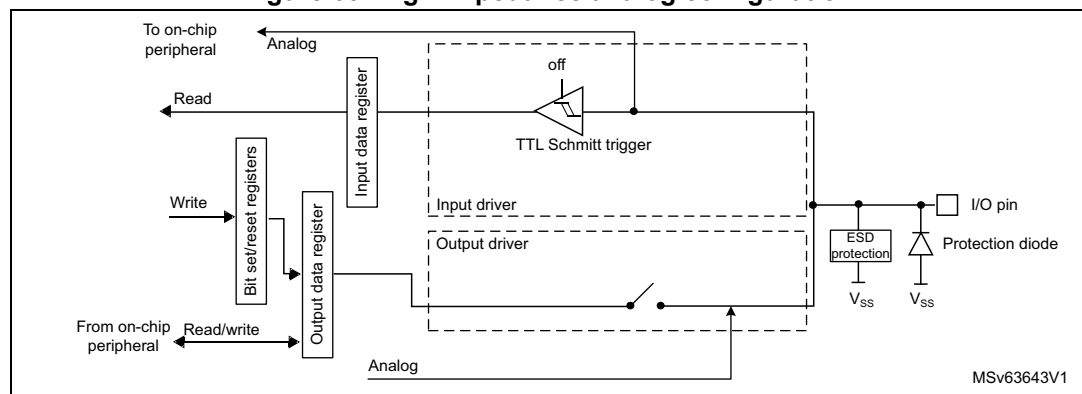
13.3.12 Analog configuration

When the I/O port is programmed as analog configuration:

- The output buffer is disabled.
- The Schmitt trigger input is deactivated, providing zero consumption for every analog value of the I/O pin. The output of the Schmitt trigger is forced to a constant value (0).
- The weak pull-up and pull-down resistors are disabled by hardware.
- Read access to the input data register gets the value 0.

Figure 63 shows the high-Z, analog-input configuration of the I/O port bits.

Figure 63. High-impedance analog configuration



13.3.13 Using the HSE or LSE oscillator pins as GPIOs

When the HSE or LSE oscillator is switched off (default state after reset), the related oscillator pins can be used as normal GPIOs.

When the HSE or LSE oscillator is switched on (by setting the HSEON or LSEON bit in the RCC_CSR register), the oscillator takes control of its associated pins and the GPIO configuration of these pins has no effect.

When the oscillator is configured in a user external clock mode, only the pin is reserved for clock input, and the OSC_OUT or OSC32_OUT pin can still be used as normal GPIO.

13.3.14 Using the GPIO pins in the RTC supply domain

The PC13/PC14/PC15/PI8 GPIO functionality is lost when the core supply domain is powered off (when the device enters Standby mode). In this case, if their GPIO configuration is not bypassed by the RTC configuration, these pins are set in an analog input mode.

For details about I/O control by the RTC, refer to [Section 46.3: RTC functional description](#)

13.3.15 I/Os state retention during standby mode

In the Standby mode, the I/Os are by default in floating state.

If the IORETEN bit in the PWR_IOPRETR register is set, the I/Os state is sampled during standby entry. The state of I/Os is applied to the pin via pull-up and pull-down resistors. The pull-up and pull-down resistors remains applied after Standby wake-up until the IORETEN bit in the PWR_IOPRETR register is cleared by software.

13.3.16 TrustZone security

The TrustZone security is activated by the TZEN option byte in the FLASH option byte register. When the TrustZone is active (TZEN = 0xB4), each I/O pin of GPIO port can be individually configured as secure through the GPIOx_SECCFGR register.

When the selected I/O pin is configured as secure, its corresponding configuration bits for alternate function, mode selection, I/O data are secure against a non-secure access. In case of non-secure access, these bits are RAZ/WI.

The I/Os with peripherals functions are also conditioned by the peripheral security configuration (see [Section 5: Global TrustZone® controller \(GTZC\)](#) for more details):

- For peripherals for which the I/O pin selection is done through alternate functions registers: if the peripheral is configured as secure, it cannot be connected to a non-secure I/O pin. If this is not respected, the input data to the secure peripheral is forced to 0 (I/O input pin value is ignored) and the output pin value is forced to 0, thus avoiding any secure information leak through non-secure I/Os.
- For I/Os with analog switches, directly controlled by peripherals (such as ADC for instance): If the I/O is secure, the I/O analog switch cannot be controlled by a non-secure peripheral. If this is not respected, the switch remains open. This prevents the redirection of secure data to a non-secure peripheral or I/O through the analog path. Refer to [Section 3: System security](#) for more details.
- Some of the paths between I/Os “additional functions” and peripherals are not blocked if the I/O is secure and the peripheral is non-secure. Therefore, it is recommended to configure those peripherals as secure even when not used by the application. Refer to [Section 3: System security](#) for the list of concerned peripherals. When the path has a security control, it follows the same rule as I/O selection through alternate functions.

Refer to the device pins definition table in datasheet for more information about peripherals alternate functions and additional functions mapping.

After reset, all GPIO ports are secure.

The table below gives a summary of the I/O port secured bits following the security configuration bit in the GPIO_SECCFGR register. When the I/O bit port is configured as secure:

- Secured bits: read and write operations are only allowed by a secure access. Non secure-read or write accesses on secured bits are RAZ/WI. There is no illegal access event generated.
- Non-secure bits: no restriction. Read and write operations are allowed by both secure and non-secure accesses.

When the TrustZone security is disabled (TZEN = 0xC3 in FLASH_OPTSR2 register), all registers bits are non-secure. The GPIOx_SECCFGR register is RAZ/WI.

Table 126. GPIO secured bits

Secure configuration bit	Secured bit	Register name	Non-secure access on secure bits
SECy = 1 in GPIOx_SECCFGR ^{(1) (2)}	MODEy[1:0]	GPIOx_MODER	RAZ/WI
	OTy	GPIOx_OTYPER	
	OSPEEDy[1:0]	GPIOx_OSPEEDR	
	PUPDy[1:0]	GPIOx_PUPDR	
	IDy	GPIOx_IDR	
	ODy	GPIOx_ODR	
	BSy and BRy	GPIOx_BSRR	
	LCKy	GPIOx_LCKR	
	BRy	GPIOx_BRR	
	AFSELy[3:0]	GPIOx_AFRH	
		GPIOx_AFRL	
	HSLVy	GPIOx_HSLVR	

1. GPIOx, x = A to I. For x = A to H, y = 0 to 15. For x = I, y = 0 to 11.

2. The number of GPIOx ports depends upon the device. Refer to the product datasheet for availability of a particular port. If not present, consider the associated bits as reserved, and keep them at reset value.

13.3.17 Privileged and unprivileged modes

All GPIO registers can be read and written by privileged and unprivileged accesses, whatever the security state (secure or non-secure).

13.3.18 High-speed low-voltage mode (HSLV)

Some I/Os have the capability to increase their maximum speed at low voltage by configuring them in HSLV mode. The I/O HSLV bit controls whether the I/O output speed is optimized to operate at 3.3 V (default setting) or at 1.8 V (HSLV = 1).

Caution: The I/O HSLV configuration bit must not be set if the I/O supply (V_{DD} or V_{DDIO2}) is above 2.7 V. Setting it while the voltage is higher than 2.7 V can damage the device. The I/O HSLV bit can be set only when the corresponding option bit is activated (IO_VDD_HSLV or IO_VDDIO2_HSLV depending on the I/O supply, refer to [Section 7.4: FLASH option bytes](#)).

There is no hardware protection associated to this feature so it is recommended to use it only as a static configuration for fixed I/O supply.

13.3.19 I/O compensation cell

The I/O commutation slew rate (tfall / trise) can be adapted by software depending on process, voltage and temperatures conditions, to reduce the I/O noise on power supply. Refer to [Section 14: System configuration, boot, and security \(SBS\)](#) for more details.

13.4 GPIO registers

This section gives a detailed description of the GPIO registers. Note that the number of GPIOx ports depends upon the device. Refer to the product datasheet for the availability of a particular port. If not present, consider the associated bits as reserved, and keep them at reset value.

The peripheral registers can be written in word, half word, or byte mode.

13.4.1 GPIO port mode register (GPIOx_MODER) (x = A to I)

Address offset: 0x00

Reset value: 0xABFF FFFF (for port A)

Reset value: 0xFFFF FEBF (for port B)

Reset value: 0xFFFF FFFF (for ports C..H)

Reset value: 0x00FF FFFF (for port I)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
MODE15[1:0]		MODE14[1:0]		MODE13[1:0]		MODE12[1:0]		MODE11[1:0]		MODE10[1:0]		MODE9[1:0]		MODE8[1:0]	
rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
MODE7[1:0]		MODE6[1:0]		MODE5[1:0]		MODE4[1:0]		MODE3[1:0]		MODE2[1:0]		MODE1[1:0]		MODE0[1:0]	
rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw

Bits 31:0 **MODEy[1:0]**: Port x configuration I/O pin y (y = 15 to 0)

These bits are written by software to configure the I/O mode.

00: Input mode

01: General purpose output mode

10: Alternate function mode

11: Analog mode (reset state)

Note: The bitfield is reserved and must be kept to reset value when the corresponding I/O is not available on the selected package.

13.4.2 GPIO port output type register (GPIOx_OTYPER) (x = A to I)

Address offset: 0x04

Reset value: 0x0000 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
OT15	OT14	OT13	OT12	OT11	OT10	OT9	OT8	OT7	OT6	OT5	OT4	OT3	OT2	OT1	OT0
rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw

Bits 31:16 Reserved, must be kept at reset value.

Bits 15:0 **OTy**: Port x configuration I/O pin y (y = 15 to 0)

These bits are written by software to configure the I/O output type.

0: Output push-pull (reset state)

1: Output open-drain

Note: The bit is reserved and must be kept to reset value when the corresponding I/O is not available on the selected package.

13.4.3 GPIO port output speed register (GPIOx_OSPEEDR) (x = A to I)

Address offset: 0x08

Reset value: 0x0C00 0000 (for port A)

Reset value: 0x0000 00C0 (for port B)

Reset value: 0x0000 0000 (for the other ports)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
OSPEED15[1:0]		OSPEED14[1:0]		OSPEED13[1:0]		OSPEED12[1:0]		OSPEED11[1:0]		OSPEED10[1:0]		OSPEED9[1:0]		OSPEED8[1:0]	
rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
OSPEED7[1:0]		OSPEED6[1:0]		OSPEED5[1:0]		OSPEED4[1:0]		OSPEED3[1:0]		OSPEED2[1:0]		OSPEED1[1:0]		OSPEED0[1:0]	
rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw

Bits 31:0 **OSPEEDy[1:0]**: Port x configuration I/O pin y (y = 15 to 0)

These bits are written by software to configure the I/O output speed.

00: Low speed

01: Medium speed

10: High speed

11: Very-high speed

Note: Refer to the device datasheet for the frequency specifications, the power supply, and the load conditions for each speed.

The bitfield is reserved and must be kept to reset value when the corresponding I/O is not available on the selected package.

13.4.4 GPIO port pull-up/pull-down register (GPIOx_PUPDR) (x = A to I)

Address offset: 0x0C

Reset value: 0x6400 0000 (for port A)

Reset value: 0x0000 0100 (for port B)

Reset value: 0x0000 0000 (for the other ports)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
PUPD15[1:0]		PUPD14[1:0]		PUPD13[1:0]		PUPD12[1:0]		PUPD11[1:0]		PUPD10[1:0]		PUPD9[1:0]		PUPD8[1:0]	
rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
PUPD7[1:0]		PUPD6[1:0]		PUPD5[1:0]		PUPD4[1:0]		PUPD3[1:0]		PUPD2[1:0]		PUPD1[1:0]		PUPD0[1:0]	
rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw

Bits 31:0 **PUPDy[1:0]**: Port x configuration I/O pin y (y = 15 to 0)

These bits are written by software to configure the I/O pull-up or pull-down

00: No pull-up, pull-down

01: Pull-up

10: Pull-down

11: Reserved

Note: The bitfield is reserved and must be kept to reset value when the corresponding I/O is not available on the selected package.

13.4.5 GPIO port input data register (GPIOx_IDR) (x = A to I)

Address offset: 0x10

Reset value: 0x0000 XXXX

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID15	ID14	ID13	ID12	ID11	ID10	ID9	ID8	ID7	ID6	ID5	ID4	ID3	ID2	ID1	ID0
r	r	r	r	r	r	r	r	r	r	r	r	r	r	r	r

Bits 31:16 Reserved, must be kept at reset value.

Bits 15:0 **IDy**: Port x input data I/O pin y (y = 15 to 0)

These bits are read-only. They contain the input value of the corresponding I/O port.

Note: The bit is reserved and must be kept to reset value when the corresponding I/O is not available on the selected package.

13.4.6 GPIO port output data register (GPIOx_ODR) (x = A to I)

Address offset: 0x14

Reset value: 0x0000 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
OD15	OD14	OD13	OD12	OD11	OD10	OD9	OD8	OD7	OD6	OD5	OD4	OD3	OD2	OD1	OD0
rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw

Bits 31:16 Reserved, must be kept at reset value.

Bits 15:0 **ODy**: Port output data I/O pin y (y = 15 to 0)

These bits can be read and written by software.

Note: For atomic bit set/reset, the OD bits can be individually set and/or reset by writing to the GPIOx_BSRR or GPIOx_BRR registers (x = A to I).

The bit is reserved and must be kept to reset value when the corresponding I/O is not available on the selected package.

13.4.7 GPIO port bit set/reset register (GPIOx_BSRR) (x = A to I)

Address offset: 0x18

Reset value: 0x0000 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
BR15	BR14	BR13	BR12	BR11	BR10	BR9	BR8	BR7	BR6	BR5	BR4	BR3	BR2	BR1	BR0
w	w	w	w	w	w	w	w	w	w	w	w	w	w	w	w
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
BS15	BS14	BS13	BS12	BS11	BS10	BS9	BS8	BS7	BS6	BS5	BS4	BS3	BS2	BS1	BS0
w	w	w	w	w	w	w	w	w	w	w	w	w	w	w	w

Bits 31:16 **BRy**: Port x reset I/O pin y (y = 15 to 0)

These bits are write-only. A read to these bits returns the value 0x0000.

0: No action on the corresponding ODy bit

1: Resets the corresponding ODy bit

Note: If both BSy and BRy are set, BSy has priority.

The bit is reserved and must be kept to reset value when the corresponding I/O is not available on the selected package.

Bits 15:0 **BSy**: Port x set I/O pin y (y = 15 to 0)

These bits are write-only. A read to these bits returns the value 0x0000.

0: No action on the corresponding ODy bit

1: Sets the corresponding ODy bit

Note: The bit is reserved and must be kept to reset value when the corresponding I/O is not available on the selected package.

13.4.8 GPIO port configuration lock register (GPIOx_LCKR) (x = A to I)

This register is used to lock the configuration of the port bits when a correct write sequence is applied to bit 16 (LCKK). The value of bits [15:0] is used to lock the configuration of the GPIO. During the write sequence, the value of LCKR[15:0] must not change. When the LOCK sequence has been applied on a port bit, the value of this port bit can no longer be modified until the next MCU reset or peripheral reset.

Note: A specific write sequence is used to write to the GPIOx_LCKR register. Only word access (32-bit long) is allowed during this locking sequence.

Each bit freezes a specific configuration register (control and alternate function registers).

Address offset: 0x1C

Reset value: 0x0000 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	LCKK
															rw
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
LCK15	LCK14	LCK13	LCK12	LCK11	LCK10	LCK9	LCK8	LCK7	LCK6	LCK5	LCK4	LCK3	LCK2	LCK1	LCK0
rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw

Bits 31:17 Reserved, must be kept at reset value.

Bit 16 **LCKK**: Lock key

This bit can be read any time. It can only be modified using the lock key write sequence.

0: Port configuration lock key not active

1: Port configuration lock key active. The GPIOx_LCKR register is locked until the next MCU reset or peripheral reset.

- LOCK key write sequence:

WR LCKR[16] = 1 + LCKR[15:0]

WR LCKR[16] = 0 + LCKR[15:0]

WR LCKR[16] = 1 + LCKR[15:0]

- LOCK key read

RD LCKR[16] = 1 (this read operation is optional but it confirms that the lock is active)

Note: During the LOCK key write sequence, the value of LCK[15:0] must not change.

Any error in the lock sequence aborts the LOCK.

After the first LOCK sequence on any bit of the port, any read access on the LCKK bit returns 1 until the next MCU reset or peripheral reset.

Bits 15:0 **LCKy**: Port x lock I/O pin y (y = 15 to 0)

These bits are read/write but can only be written when the LCKK bit is 0

0: Port configuration not locked

1: Port configuration locked

Note: The bit is reserved and must be kept to reset value when the corresponding I/O is not available on the selected package.

13.4.9 GPIO alternate function low register (GPIOx_AFRL) (x = A to I)

Address offset: 0x20

Reset value: 0x0000 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
AFSEL7[3:0]				AFSEL6[3:0]				AFSEL5[3:0]				AFSEL4[3:0]			
rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
AFSEL3[3:0]				AFSEL2[3:0]				AFSEL1[3:0]				AFSEL0[3:0]			
rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw

Bits 31:0 **AFSELY[3:0]**: Alternate function selection for port x I/O pin y (y = 7 to 0)

These bits are written by software to configure alternate function I/Os.

- 0000: AF0
- 0001: AF1
- 0010: AF2
- 0011: AF3
- 0100: AF4
- 0101: AF5
- 0110: AF6
- 0111: AF7
- 1000: AF8
- 1001: AF9
- 1010: AF10
- 1011: AF11
- 1100: AF12
- 1101: AF13
- 1110: AF14
- 1111: AF15

Note: The bitfield is reserved and must be kept to reset value when the corresponding I/O is not available on the selected package.

13.4.10 GPIO alternate function high register (GPIOx_AFRH) (x = A to H)

Address offset: 0x24

Reset value: 0x0000 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
AFSEL15[3:0]				AFSEL14[3:0]				AFSEL13[3:0]				AFSEL12[3:0]			
rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
AFSEL11[3:0]				AFSEL10[3:0]				AFSEL9[3:0]				AFSEL8[3:0]			
rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw

Bits 31:0 **AFSELY[3:0]**: Alternate function selection for port x I/O pin y (y = 15 to 8)

These bits are written by software to configure alternate function I/Os.

0000: AF0
 0001: AF1
 0010: AF2
 0011: AF3
 0100: AF4
 0101: AF5
 0110: AF6
 0111: AF7
 1000: AF8
 1001: AF9
 1010: AF10
 1011: AF11
 1100: AF12
 1101: AF13
 1110: AF14
 1111: AF15

Note: The bitfield is reserved and must be kept to reset value when the corresponding I/O is not available on the selected package.

13.4.11 GPIO port bit reset register (GPIOx_BRR) (x = A to I)

Address offset: 0x28

Reset value: 0x0000 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
BR15	BR14	BR13	BR12	BR11	BR10	BR9	BR8	BR7	BR6	BR5	BR4	BR3	BR2	BR1	BR0
w	w	w	w	w	w	w	w	w	w	w	w	w	w	w	w

Bits 31:16 Reserved, must be kept at reset value.

Bits 15:0 **BRy**: Port x reset IO pin y (y = 15 to 0)

These bits are write-only. A read to these bits returns the value 0x0000.

0: No action on the corresponding ODy bit

1: Reset the corresponding ODy bit

Note: The bit is reserved and must be kept to reset value when the corresponding I/O is not available on the selected package.

13.4.12 GPIO high-speed low-voltage register (GPIOx_HSLVR) (x = A to I)

Address offset: 0x2C

Reset value: 0x0000 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
HSLV 15	HSLV 14	HSLV 13	HSLV 12	HSLV 11	HSLV 10	HSLV 9	HSLV 8	HSLV 7	HSLV 6	HSLV 5	HSLV 4	HSLV 3	HSLV 2	HSLV 1	HSLV 0
rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw

Bits 31:16 Reserved, must be kept at reset value.

Bits 15:0 **HSLVy**: Port x high-speed low-voltage configuration (y = 15 to 0)

These bits are written by software to optimize the I/O speed when the I/O supply is low.

Each bit is active only if the corresponding IO_VDD_HSLV/IO_VDDIO2_HSLV user option bit is set. It must be used only if the I/O supply voltage is below 2.7 V.

Setting these bits when the I/O supply (VDD or VDDIO2) is higher than 2.7 V may be destructive.

0: I/O speed optimization disabled

1: I/O speed optimization enabled

Note: Not all I/Os support the HSLV mode. Refer to the I/O structure in the corresponding datasheet for the list of I/Os supporting this feature. Other I/Os HSLV configuration must be kept at reset value.

The bit is reserved and must be kept to reset value when the corresponding I/O is not available on the selected package.

13.4.13 GPIO secure configuration register (GPIOx_SECCFGR) (x = A to I)

When the system is secure (TZEN = 0xB4), this register provides write access security and can be written only by a secure access. It is used to configure a selected I/O as secure. A non-secure write access to this register is discarded.

When the system is not secure (TZEN = 0xC3), this register is RAZ/WI.

Address offset: 0x30

Reset value: 0x0000 FFFF (for ports A to H)

Reset value: 0x0000 0FFF (for port I)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
SEC15	SEC14	SEC13	SEC12	SEC11	SEC10	SEC9	SEC8	SEC7	SEC6	SEC5	SEC4	SEC3	SEC2	SEC1	SEC0
rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw

Bits 31:16 Reserved, must be kept at reset value.

Bits 15:0 **SECy**: I/O pin of Port x secure bit enable y (y = 15 to 0)

These bits are written by software to enable or disable the I/O port pin security.

0: The I/O pin is non-secure

1: The I/O pin is secure. Refer to [Table 126](#) for all corresponding secured bits.

Note: The bit is reserved and must be kept to reset value when the corresponding I/O is not available on the selected package.

13.4.14 GPIO register map

Table 127. GPIO register map and reset values

Offset	Register	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0																																
0x00	GPIOx_MODER (x = A to I)	MODE15[1:0]				MODE14[1:0]				MODE13[1:0]				MODE12[1:0]				MODE11[1:0]				MODE10[1:0]				MODE9[1:0]				MODE8[1:0]				MODE7[1:0]				MODE6[1:0]				MODE5[1:0]				MODE4[1:0]				MODE3[1:0]				MODE2[1:0]				MODE1[1:0]				MODE0[1:0]			
	Reset value for port A	1	0	1	0	1	0	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1															
	Reset value for port B	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	0	1	0	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1															
	Reset value for ports C...H	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1															
	Reset value for port I	0	0	0	0	0	0	0	0	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1																
0x04	GPIOx_OTYPER (x = A to I)	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	OT15	OT14	OT13	OT12	OT11	OT10	OT9	OT8	OT7	OT6	OT5	OT4	OT3	OT2	OT1	OT0																																
	Reset value																	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0																
0x08	GPIOx_OSPEEDR (x = A to I)	OSPEED15[1:0]				OSPEED14[1:0]				OSPEED13[1:0]				OSPEED12[1:0]				OSPEED11[1:0]				OSPEED10[1:0]				OSPEED9[1:0]				OSPEED8[1:0]				OSPEED7[1:0]				OSPEED6[1:0]				OSPEED5[1:0]				OSPEED4[1:0]				OSPEED3[1:0]				OSPEED2[1:0]				OSPEED1[1:0]				OSPEED0[1:0]			
	Reset value for port A	0	0	0	0	1	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0																
	Reset value for port B	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0																
	Reset value for ports C...I	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0																
0x0C	GPIOx_PUPDR (x = A to I)	PUPD15[1:0]				PUPD14[1:0]				PUPD13[1:0]				PUPD12[1:0]				PUPD11[1:0]				PUPD10[1:0]				PUPD9[1:0]				PUPD8[1:0]				PUPD7[1:0]				PUPD6[1:0]				PUPD5[1:0]				PUPD4[1:0]				PUPD3[1:0]				PUPD2[1:0]				PUPD1[1:0]				PUPD0[1:0]			
	Reset value for port A	0	1	1	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0																
	Reset value for port B	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0																
	Reset value for ports C...I	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0																
0x10	GPIOx_IDR (x = A to I)	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	ID15	ID14	ID13	ID12	ID11	ID10	ID9	ID8	ID7	ID6	ID5	ID4	ID3	ID2	ID1	ID0																																
	Reset value																	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X														
0x14	GPIOx_ODR (x = A to I)	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	OD15	OD14	OD13	OD12	OD11	OD10	OD9	OD8	OD7	OD6	OD5	OD4	OD3	OD2	OD1	OD0																																
	Reset value																	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0												
0x18	GPIOx_BSRR (x = A to I)	BR15	BR14	BR13	BR12	BR11	BR10	BR9	BR8	BR7	BR6	BR5	BR4	BR3	BR2	BR1	BR0	BS15	BS14	BS13	BS12	BS11	BS10	BS9	BS8	BS7	BS6	BS5	BS4	BS3	BS2	BS1	BS0																																
	Reset value	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0																											

Table 127. GPIO register map and reset values (continued)

Offset	Register	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
0x28	GPIOx_BRR (x = A to I)	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	BR15	BR14	BR13	BR12	BR11	BR10	BR9	BR8	BR7	BR6	BR5	BR4	BR3	BR2	BR1	BR0
	Reset value																	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
0x2C	GPIOx_HSLVR (x = A to I)	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	HSLV15	HSLV14	HSLV13	HSLV12	HSLV11	HSLV10	HSLV9	HSLV8	HSLV7	HSLV6	HSLV5	HSLV4	HSLV3	HSLV2	HSLV1	HSLV0
	Reset value																	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
0x30	GPIOx_SECCFGR (x = A to I)	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	SEC15	SEC14	SEC13	SEC12	SEC11	SEC10	SEC9	SEC8	SEC7	SEC6	SEC5	SEC4	SEC3	SEC2	SEC1	SEC0
	Reset value for A to H																	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
	Reset value for port I																	0	0	0	0	1	1	1	1	1	1	1	1	1	1	1	1

Refer to [Section 2.3](#) for the register boundary addresses.